Software Requirements Specification

Phased Roadmap — Billing & POS Platform for Retail and Restaurant Chains

Phase 1: Core Billing (Uploaded Stock) → **Phase 2: Vendor & Stock Maintenance** → **Phase 3: Multi-Outlet & Chains**

Version 2.0 (Phased Revision)

Prepared: September 2026

Status: Draft for Review

Document Approval

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for a Billing and Point-of-Sale (POS) platform for retail stores and restaurants, delivered as three incremental phases. Each phase is scoped to be a complete, independently usable product — not a partial build that only becomes useful once a later phase ships.

1.2 Phased Delivery Approach

Rather than building the full multi-outlet, multi-format platform in one release, the product is delivered in three phases of increasing scope:

- Phase 1 proves the core billing loop for one store, with stock loaded by upload rather than through a purchasing workflow.
- Phase 2 adds the procurement side — vendor management, purchase orders and ongoing stock maintenance — so the same store no longer depends on manual re-uploads to keep stock accurate.
- Phase 3 scales the proven single-store product out to multiple outlets and franchises, and extends it to cover restaurant/QSR operations, loyalty, and third-party integrations — turning it into a full retail-and-restaurant chain platform.

This staged approach lets a pilot store go live quickly on Phase 1, validates the product with real usage before procurement complexity is added, and defers the highest-complexity, chain-scale features until the core transaction engine is proven.

1.3 Document Conventions

Requirement identifiers follow the pattern FR-P<phase>-<MODULE>-<NUMBER> for functional requirements and NFR-P<phase>-<CATEGORY>-<NUMBER> for non-functional requirements, so every requirement is traceable to the phase that introduces it. Priority (High/Medium/Low) is scoped within its own phase: a High item is necessary for that phase's release, not necessarily for the end-state product.

1.4 Intended Audience

- Product managers and business analysts sequencing scope and release plans
- UX/UI designers and software architects
- Development and QA teams responsible for build and test, phase by phase
- Retail/restaurant business owners and operations stakeholders reviewing what ships when
- Implementation and support teams during phased rollout

1.5 Product Scope (End State)

By the end of Phase 3, the product is a cloud-connected, offline-capable billing and retail/restaurant management platform covering point-of-sale billing, inventory and purchase management, restaurant table/kitchen operations, customer loyalty, multi-outlet/franchise administration, taxation and statutory compliance, payments, and business reporting/analytics — comparable in scope to established retail/restaurant ERP-POS suites such as Zoho and GoFrugal.

Out of scope for all phases of this version: full HR/payroll processing (only data export is covered in Phase 3), advanced supply-chain/warehouse-management-system functionality beyond store-level inventory, and native support for industries outside general retail and food service.

1.6 Definitions, Acronyms & Abbreviations

Term	Definition
POS	Point of Sale — the checkout terminal/software used to process a sale.
SKU	Stock Keeping Unit — a unique identifier for a sellable item/variant.
GRN	Goods Receipt Note — document confirming receipt of goods against a purchase order.
KOT / KDS	Kitchen Order Ticket / Kitchen Display System — order routing to the kitchen, paper or digital.
HSN/SAC	Tax classification codes used for GST computation.
IRN	Invoice Reference Number issued by the tax portal for an e-invoice.
HO	Head Office — the central administrative entity overseeing multiple outlets.
Z-Report	End-of-day summary report of sales, tax and payment totals.
RPO / RTO	Recovery Point / Time Objective — disaster-recovery targets.
MVP	Minimum Viable Product.

1.7 References

- GST/e-Invoicing and e-Way Bill statutory portal specifications (jurisdiction-specific)
- PCI-DSS Data Security Standard
- ISO/IEC/IEEE 29148 guidance for software requirements documentation
- Comparable market products: Zoho Books/Inventory/POS suite, GoFrugal RetailEasy/ServeEasy

2. Roadmap Overview

2.1 Phase Summary

Phase	Theme	Outlet Scope	Key Capabilities Added	What Makes It Usable Alone
Phase 1	Core Billing, Uploaded Stock	Single store	Fast checkout, tax, payments, receipts, essential reports, stock via bulk upload	A store can fully replace manual/cash-register billing on day one.
Phase 2	Vendor & Stock Maintenance	Single store	Purchase orders, GRN, vendor ledger, stock adjustments, batch/expiry, reorder alerts	The same store can now run its full buy-side stock lifecycle without spreadsheets.
Phase 3	Multi-Outlet, Franchise & Chains	Unlimited outlets, retail + restaurant	HO console, franchise/royalty, restaurant/KOT/KDS, loyalty/CRM, aggregator & accounting integrations, e-invoicing, multi-outlet sync	An operator can run a retail or restaurant chain, company-owned or franchised, from one platform.

2.2 Cumulative Scope Matrix

" " marks capability present as of that phase; "—" marks not yet introduced; "+" marks what is newly added in that phase on top of the previous one.

Capability Area	Phase 1	Phase 2	Phase 3
Billing & checkout	(core)	(unchanged)	(extended: gateway integration, reconciliation)
Stock entry	Bulk upload only	+ Purchase orders, GRN, adjustments	+ Multi-outlet stock transfer
Vendor/procurement	—		(unchanged)
Tax & compliance	Basic tax calculation	(unchanged)	+ E-invoicing, e-way bill, multi-jurisdiction
Reporting	Daily Z-report, item sales	+ Purchase, valuation, vendor reports	+ Consolidated BI, custom builder, drill-down
Outlets	1 (fixed)	1 (fixed)	Unlimited, with HO console
Franchise management	—	—	
Restaurant / QSR ops	—	—	
Loyalty & CRM	—	—	
Delivery aggregator / e-commerce	—	—	
Accounting integration	—	—	

3-. Overall Description

3.1 Product Perspective

Phase 1 and Phase 2 are delivered as a self-contained, single-store application requiring no central cloud tenant or Head Office console. Phase 3 evolves the same data model into a multi-tenant, cloud-connected platform with a Head Office console, outlet-level POS/restaurant clients, and an integration layer for payments, tax portals, accounting and delivery aggregators. Each phase's data model is designed to extend into the next without a rewrite.

3.2 User Classes Across Phases

User Class	Introduced In	Needs
Cashier	Phase 1	Fast, low-error, minimal-training billing screen.
Admin (Store Owner/Manager)	Phase 1	Item/stock upload, tax and store configuration, reports, backup.
Purchasing/Inventory Staff	Phase 2	Vendor management, purchase orders, goods receipt, stock adjustments.
Waiter / Captain	Phase 3	Mobile, tableside order entry synced to kitchen and bill.
Kitchen Staff	Phase 3	Clear, prioritized view of pending orders (KOT/KDS).
Head Office Admin	Phase 3	Network-wide visibility, master-data control, consolidated analytics.
Accountant / Finance	Phase 3	Statutory reports and accounting-system sync across outlets.

3.3 Operating Environment Evolution

- Phase 1–2: single Windows PC or Android tablet at the store counter; local database; internet used only for backup and optional updates.
- Phase 3: cloud-hosted multi-tenant back end; POS clients on Windows/Android, restaurant/captain apps on Android/iOS, HO console on standard web browsers.

3.4 Assumptions and Dependencies

- A Phase 1/2 pilot store operates from a single physical location with one tax jurisdiction.
- Hardware (printers, scanners, cash drawers) is procured to platform-certified specifications from Phase 1 onward.
- Phase 3's third-party integrations (payment gateways, tax portals, aggregators) depend on stable, documented external APIs.
- Migrating a Phase 1/2 store's data into the Phase 3 multi-outlet model is assumed to use the migration tooling defined in Phase 3 (see FR-P3-ADM-004).

4. Phase 1 — Core Billing with Uploaded Stock (Single Store)

4.1 Phase Goal

Ship the smallest reliable product that lets one store fully replace manual or cash-register billing: fast checkout, correct tax, and a daily sales report — with stock loaded by upload rather than through a purchasing workflow.

Usable product at end of this phase: a single store can open, bill customers all day using items and quantities loaded via a stock upload, accept cash/card/UPI payment, print compliant receipts, and close the day with an accurate sales and cash report — with no dependency on Phase 2 or 3 features.

4.2 Functional Requirements

4.2.1 User Access & Login

ID	Requirement	Description	Priority
FR-P1-UM-001	Two Fixed Roles	The system shall support exactly two roles: Admin (full access) and Cashier (billing screen access only).	High
FR-P1-UM-002	PIN/Password Login	The system shall require a username/PIN or password login before any billing or admin action.	High
FR-P1-UM-003	Basic Action Log	The system shall log who created, edited or cancelled a bill, with timestamp.	Medium
FR-P1-UM-004	Auto-Lock	The screen shall lock after a configurable idle period and require re-login.	Low

4.2.2 Stock Data (Upload-Based)

Phase 1 deliberately excludes any purchase/procurement workflow. Stock enters the system only via bulk upload; ongoing procurement-driven stock maintenance is delivered in Phase 2.

ID	Requirement	Description	Priority
FR-P1-STK-001	Bulk Item & Stock Upload	The Admin shall be able to bulk-upload the item master and opening stock via a CSV/Excel template (SKU, barcode, name, category, cost price, selling price, tax rate, opening quantity).	High
FR-P1-STK-002	Single Item Quick-Add	The Admin shall be able to add or edit one item's core details and stock quantity directly, for items not covered by a bulk upload.	High
FR-P1-STK-003	Automatic Stock Deduction	The system shall automatically reduce stock on every completed sale and restore it on a valid return.	High
FR-P1-STK-004	Stock-on-Hand Visibility	The system shall provide a current stock-on-hand view/report per item.	High
FR-P1-STK-005	Low-Stock Indicator (Read-Only)	The system shall visually flag items at or below a configured minimum stock level. No reorder workflow is included in Phase 1.	Medium
FR-P1-STK-006	Stock Refresh via Re-Upload	The Admin shall be able to re-upload the stock file to correct or top up quantities, with a preview of changes before committing.	High

4.2.3 Point-of-Sale Billing (Core)

ID	Requirement	Description	Priority
FR-P1-POS-001	Barcode / Search Billing	The system shall allow adding items to a bill by barcode scan or name/code search.	High
FR-P1-POS-002	Quantity & Price Edit	The cashier shall be able to edit line quantity and, if authorized, override the unit price.	High
FR-P1-POS-003	Bill-Level Discount	The system shall support a manual flat or percentage discount at the bill level, restricted to Admin approval above a configured limit.	High
FR-P1-POS-004	Multiple Price Lists	The system shall support MRP and wholesale price lists selectable at billing time.	Medium
FR-P1-POS-005	Multi-Mode & Split Payment	The system shall accept cash, card and UPI, including split payment across two modes in a single bill.	High
FR-P1-POS-006	Change Calculation	For cash payments, the system shall calculate and display change due.	High
FR-P1-POS-007	Receipt Printing	The system shall print a thermal receipt showing itemized bill, tax breakup, payment mode and store details, with an optional SMS/email digital copy.	High
FR-P1-POS-008	Returns & Exchanges	The system shall support full or partial returns against a bill number, reversing stock and recording a refund.	Medium
FR-P1-POS-009	Hold & Recall Bill	The cashier shall be able to park an in-progress bill and resume it later.	Low
FR-P1-POS-010	Sequential Invoice Numbering	The system shall generate a unique, sequential invoice number per financial year.	High

4.2.4 Tax Configuration

ID	Requirement	Description	Priority
FR-P1-TAX-001	Tax Rate Configuration	The Admin shall be able to define one or more tax rates (e.g., GST slabs) and assign them to items.	High
FR-P1-TAX-002	Tax-Inclusive/Exclusive Pricing	The system shall support configuring whether item prices are entered tax-inclusive or tax-exclusive.	Medium
FR-P1-TAX-003	Daily Tax Summary	The system shall provide a report summarizing tax collected, grouped by tax rate, for a selected date range.	Medium

4.2.5 Essential Reporting

ID	Requirement	Description	Priority
FR-P1-RPT-001	Daily Sales (Z) Report	The system shall generate an end-of-day summary of total sales, tax, discounts and payment-mode breakup.	High
FR-P1-RPT-002	Sales by Item Report	The system shall report quantity and value sold per item over a selected date range.	Medium
FR-P1-RPT-003	Cash Drawer Report	The system shall show expected vs. counted cash at shift/day close.	Medium
FR-P1-RPT-004	Stock-on-Hand Report	The system shall provide a report of current stock quantity and value by item.	High

4.2.6 Hardware Integration

ID	Requirement	Description	Priority
FR-P1-HW-001	Core Peripherals	The system shall support a thermal receipt printer, barcode scanner and cash drawer.	High
FR-P1-HW-002	Weighing Scale Integration	The system shall optionally integrate with an electronic weighing scale for loose/weighed items.	Low

4.2.7 Store Settings & Administration

ID	Requirement	Description	Priority
FR-P1-ADM-001	Store Profile	The Admin shall be able to set store name, address, contact and tax ID for printing on receipts.	High
FR-P1-ADM-002	Data Backup	The system shall support manual and scheduled local/cloud backup of the store database.	High
FR-P1-ADM-003	Printer & Scanner Setup	The Admin shall be able to configure the connected receipt printer and barcode scanner.	High

4.3 Non-Functional Requirements

4.3.1 Performance

ID	Requirement	Description	Priority
NFR-P1-PERF-001	Billing Speed	A line item shall be added to the bill within 1 second of a barcode scan.	High
NFR-P1-PERF-002	Bill Completion Time	A typical 10-item bill shall be completed, printed and closed within 60 seconds of cashier operation.	Medium

4.3.2 Reliability & Data Integrity

ID	Requirement	Description	Priority
NFR-P1-REL-001	No Data Loss on Crash	An application crash or power loss shall not corrupt already-saved bills or stock data.	High
NFR-P1-REL-002	Automatic Local Save	The system shall persist an in-progress bill locally so it can be recovered after an unexpected restart.	Medium

4.3.3 Security

ID	Requirement	Description	Priority
NFR-P1-SEC-001	Credential Protection	Login PINs/passwords shall be stored using standard hashing, never in plain text.	High
NFR-P1-SEC-002	Local Data Protection	The store database shall not be readable/editable outside the application without authorized access.	Medium

4.3.4 Usability & Portability

ID	Requirement	Description	Priority
NFR-P1-USE-001	Minimal Training	A new cashier shall be able to complete a basic sale unaided within 15 minutes of first use.	High
NFR-P1-USE-	Single-Device Operation	The application shall run on a single Windows PC or	High

002		Android tablet without requiring a server.	
NFR-P1-USE-003	No Mandatory Internet	Core billing shall function without an active internet connection; internet is required only for backup and optional updates.	High

4.4 Explicitly Deferred Beyond Phase 1

- Vendor management, purchase orders, goods receipt and any procurement-driven stock entry (Phase 2)
- Stock adjustments, batch/expiry tracking, reorder suggestions and cycle-count audits (Phase 2)
- Multiple outlets, franchise structures, and a Head Office console (Phase 3)
- Restaurant/QSR operations — tables, KOT, KDS, menus (Phase 3)
- Loyalty programs, coupons, CRM campaigns (Phase 3)
- Delivery-aggregator, e-commerce, accounting-software and payment-gateway integrations (Phase 3)
- E-invoicing, e-way bill and multi-jurisdiction tax templates (Phase 3)

5. Phase 2 — Vendor Management, Purchase Orders & Stock Maintenance

5.1 Phase Goal

Extend the Phase 1 store with a full procurement lifecycle, so stock accuracy no longer depends on periodic manual re-uploads. All Phase 1 capabilities remain in place and continue to operate unchanged.

Usable product at end of this phase: the same single store now manages its complete buy-side workflow — raising purchase orders, receiving goods, tracking vendor payables, adjusting for damage/loss, and monitoring batches and expiry — without spreadsheets or manual stock file uploads, while billing continues exactly as in Phase 1.

5.2 Functional Requirements

5.2.1 Vendor & Purchase Management

Introduces the procurement lifecycle that Phase 1 deliberately left out, so stock no longer depends solely on bulk re-uploads.

ID	Requirement	Description	Priority
FR-P2-PUR-001	Vendor Master	The system shall maintain vendor records with contact details and payment terms.	High
FR-P2-PUR-002	Purchase Order Creation & Approval	The Admin shall be able to create a purchase order and route it through an approval step before it is sent to the vendor.	High
FR-P2-PUR-003	Goods Receipt (GRN) Against PO	The system shall support recording goods received against an open purchase order, updating stock and flagging quantity variances.	High
FR-P2-PUR-004	Direct / Ad-hoc Stock Receipt	The system shall support recording stock received without a prior purchase order, for walk-in or informal supply.	Medium
FR-P2-PUR-005	Purchase Returns	The system shall support returning goods to a vendor with automatic debit-note generation.	Medium
FR-P2-PUR-006	Vendor Ledger / Payables	The system shall maintain a running payable balance per vendor based on purchases and payments recorded.	Medium
FR-P2-PUR-007	Landed Cost Allocation	The system shall allow allocation of freight and incidental charges into item landed cost.	Low

5.2.2 Stock Maintenance

Extends Phase 1's upload-only stock model with ongoing, operational stock-keeping controls.

ID	Requirement	Description	Priority
FR-P2-STK-001	Stock Adjustment	The system shall allow manual stock correction for damage, loss or theft, with a mandatory reason and Admin approval.	High
FR-P2-STK-002	Batch & Expiry Tracking	The system shall support batch numbers and expiry dates, with FEFO/FIFO-based issue for perishable goods.	Medium
FR-P2-STK-	Reorder Alerts & Purchase	The system shall generate reorder alerts and suggested	Medium

003	Suggestions	purchase quantities based on configured minimum stock levels and sales history.	
FR-P2-STK-004	Physical Stock Audit	The system shall support a physical stock count (manual or barcode scan) with automatic variance reporting against book stock.	Medium

5.2.3 Extended Reporting

ID	Requirement	Description	Priority
FR-P2-RPT-001	Purchase Register	The system shall report all purchases over a selected date range, by vendor and by item.	Medium
FR-P2-RPT-002	Vendor Payables Report	The system shall report outstanding payable balances by vendor.	Medium
FR-P2-RPT-003	Stock Valuation Report	The system shall report the current value of stock on hand, based on cost price.	Medium
FR-P2-RPT-004	Batch/Expiry Report	The system shall report stock nearing expiry within a configurable window.	Low

5.3 Non-Functional Requirements

5.3.1 Data Integrity & Security

ID	Requirement	Description	Priority
NFR-P2-001	Concurrent Stock Update Safety	The system shall prevent stock-count race conditions when a sale and a goods receipt are recorded at, or near, the same time.	High
NFR-P2-002	Approval Audit Trail	The system shall retain an audit trail of who approved each purchase order, stock adjustment and purchase return.	Medium

5.3.2 Performance

ID	Requirement	Description	Priority
NFR-P2-PERF-001	Procurement Workflow Time	Creating a purchase order and its corresponding goods receipt shall each complete in under 5 seconds of processing time.	Medium

5.4 Explicitly Deferred Beyond Phase 2

- Multiple outlets, stock transfers between locations, and franchise structures (Phase 3)
- Head Office console, centralized master-data push and royalty computation (Phase 3)
- Restaurant/QSR operations — tables, KOT, KDS, menus (Phase 3)
- Loyalty programs, coupons, CRM campaigns (Phase 3)
- Delivery-aggregator, e-commerce, accounting-software and payment-gateway integrations (Phase 3)
- E-invoicing, e-way bill and multi-jurisdiction tax templates (Phase 3)

6. Phase 3 — Multi-Outlet, Franchise & Retail/Restaurant Chain Expansion

6.1 Phase Goal

Scale the proven single-store product (Phases 1–2) out to unlimited outlets under centralized or franchised ownership, and extend it to cover restaurant/QSR operations — turning it into a full retail-and-restaurant chain platform comparable to Zoho and GoFrugal.

Usable product at end of this phase: an operator can run a multi-outlet retail chain, a franchised network, or a restaurant/QSR chain — or a mix of formats — from one Head Office console, with centralized pricing, consolidated reporting, restaurant table/kitchen operations, loyalty, and integrations to payments, accounting and delivery aggregators.

6.2 Functional Requirements

6.2.1 Store, Outlet & Franchise Management

ID	Requirement	Description	Priority
FR-P3-STR-001	Multi-Outlet Setup	The Head Office shall be able to create, configure, activate and deactivate unlimited outlets/branches from a central console.	High
FR-P3-STR-002	Franchise Hierarchy	The system shall classify outlets as company-owned or franchised, with separate billing, royalty and reporting rules per class.	High
FR-P3-STR-003	Outlet-Specific Configuration	Each outlet shall support independent tax rates, price lists and operating hours while inheriting central master data.	High
FR-P3-STR-004	Centralized Master Data & Push	Product, pricing and tax masters shall be maintained centrally and pushed to selected outlets or the entire network.	High
FR-P3-STR-005	Stock Transfer Between Outlets	The system shall support a stock-transfer workflow, with in-transit tracking, between outlets/warehouses.	High
FR-P3-STR-006	Outlet Performance Comparison	The HO dashboard shall provide side-by-side comparison of sales, footfall and average bill value across outlets.	Medium

6.2.2 Head Office & Franchise Console

ID	Requirement	Description	Priority
FR-P3-HO-001	Central Monitoring	The HO console shall provide live visibility into all outlets' sales, stock and cash positions from a single screen.	High
FR-P3-HO-002	Price & Promotion Push	The HO shall be able to push price changes and promotions to selected outlets or the entire network simultaneously.	High
FR-P3-HO-003	Franchise Royalty Computation	The system shall auto-compute franchise royalty/revenue-share based on configurable rules and generate royalty invoices.	Medium
FR-P3-HO-	Exception Approval	High-value discounts, refunds and stock write-offs shall	Medium

004	Workflow	be routed through an HO/manager approval workflow.	

6.2.3 Restaurant / QSR Operations

Extends the platform, first proven on retail billing, to dine-in and quick-service food operations for restaurant chains.

ID	Requirement	Description	Priority
FR-P3-RES-001	Visual Table Management	The system shall provide a floor-plan view showing table status (vacant, occupied, reserved, billed) with drag-and-drop transfer/merge.	High
FR-P3-RES-002	Kitchen Order Ticket (KOT)	The system shall auto-generate KOTs routed to the correct kitchen/bar printer or station based on item category.	High
FR-P3-RES-003	Kitchen Display System (KDS)	The system shall support digital KDS screens showing a live order queue with preparation status.	Medium
FR-P3-RES-004	Order Type Handling	The system shall support Dine-in, Takeaway, Delivery and Drive-thru order modes, each with an appropriate workflow.	High
FR-P3-RES-005	Menu & Recipe Management	The system shall support menu categories, modifiers/add-ons, combo deals, time-based availability, and recipe-based (BOM) ingredient deduction.	High
FR-P3-RES-006	Captain/Waiter Order App	The system shall provide a mobile app for tableside order-taking that syncs instantly to the kitchen and billing engine.	Medium
FR-P3-RES-007	Flexible Bill Settlement	The system shall support table-wise, guest-wise (split-by-head) and item-wise bill settlement.	High

6.2.4 Pricing, Promotions & Loyalty

ID	Requirement	Description	Priority
FR-P3-PROMO-001	Advanced Discount Rules Engine	The system shall support item-, bill-, category- and customer-group-level discount rules with an approval workflow.	Medium
FR-P3-PROMO-002	Coupons & Gift Vouchers	The system shall support creation, redemption and tracking of coupons and gift vouchers.	Medium
FR-P3-PROMO-003	Loyalty Program	The system shall support point- or tier-based loyalty programs with configurable earn/redeem rules across outlets.	High
FR-P3-PROMO-004	Combo & Time-Based Offers	The system shall support bundle pricing and time-bound offers such as happy-hour discounts.	Medium

6.2.5 Customer Relationship Management

ID	Requirement	Description	Priority
FR-P3-CRM-001	Unified Customer Database	The system shall maintain a single customer master with consolidated purchase history across all outlets.	High
FR-P3-CRM-002	Lifecycle Campaigns	The system shall support automated SMS/email/WhatsApp campaigns triggered by events such as birthdays.	Low
FR-P3-CRM-	Customer Segmentation	The system shall segment customers by recency,	Low

003		frequency and monetary (RFM) value for targeted marketing.	

6.2.6 Delivery Aggregator & Online Ordering

ID	Requirement	Description	Priority
FR-P3-DLV-001	Aggregator Order Ingestion	The system shall integrate with food-delivery aggregator platforms to receive orders directly into the POS/KOT workflow.	High
FR-P3-DLV-002	Menu & Availability Sync	The system shall push menu, price and item-availability updates to all connected aggregator platforms.	Medium
FR-P3-DLV-003	Branded Online Ordering	The system shall support a branded online-ordering channel integrated with real-time inventory and order status.	Low

6.2.7 Accounting & Third-Party Integration

ID	Requirement	Description	Priority
FR-P3-INT-001	Accounting Software Sync	The system shall integrate with accounting platforms (e.g., Tally, QuickBooks, Zoho Books) to auto-post sales, purchase and expense entries.	High
FR-P3-INT-002	Payment Gateway Integration	The system shall integrate with standard payment gateways/UPI QR providers for real-time confirmation and automated settlement reconciliation.	High
FR-P3-INT-003	Open API & Webhooks	The system shall expose documented REST APIs and webhooks for custom third-party integrations.	Medium
FR-P3-INT-004	E-Commerce Marketplace Sync	The system shall integrate with e-commerce platforms for unified inventory and order management.	Low

6.2.8 Extended Tax & Regulatory Compliance

ID	Requirement	Description	Priority
FR-P3-TAX-001	E-Invoicing	The system shall generate IRN- and QR-code-compliant e-invoices and support real-time reporting to the applicable tax portal.	High
FR-P3-TAX-002	E-Way Bill Generation	The system shall generate e-way bills for stock transfers and sales above the statutory value threshold.	Medium
FR-P3-TAX-003	Multi-Jurisdiction Tax Templates	The system shall support configurable tax templates for outlets spanning multiple states or countries.	Medium

6.2.9 Multi-Outlet Offline Mode & Sync

ID	Requirement	Description	Priority
FR-P3-OFF-001	Offline Billing Continuity	Each outlet's POS shall continue billing, KOT generation and payment capture without an active internet connection.	High
FR-P3-OFF-002	Auto-Sync on Reconnect	Offline transactions shall automatically queue and synchronize to the central server on reconnection, applying defined conflict-resolution rules.	High

6.2.10 Notifications, Multi-Language & Data Migration

ID	Requirement	Description	Priority
FR-P3-ADM-001	Operational Alerts	The system shall send configurable alerts (low stock, cash variance, large discount) via app notification, SMS or email.	Medium
FR-P3-ADM-002	Customer Order Notifications	The system shall notify customers of order status via SMS/WhatsApp for delivery and takeaway orders.	Low
FR-P3-ADM-003	Multi-Language & Multi-Currency	The system shall support the UI, receipts and reporting in multiple regional languages and currencies for international operators.	Low
FR-P3-ADM-004	Bulk Data Migration Tools	The system shall provide tools/templates to migrate historical item, customer and transaction data from Phase 1/2 or legacy systems into the multi-outlet data model.	High

6.2.11 Extended Access Control & Reporting

ID	Requirement	Description	Priority
FR-P3-UM-001	Full Role-Based Access Control	The system shall support configurable roles (Store Manager, Waiter, Kitchen Staff, Accountant, HO Admin) with granular, module-level permissions.	High
FR-P3-UM-002	Custom Report Builder	Authorized users shall be able to build and save custom report templates across outlets.	Medium
FR-P3-UM-003	Drill-Down Analytics	Consolidated all-outlet dashboards shall support drill-down to outlet, terminal and individual transaction level.	Medium

6.3 Non-Functional Requirements

6.3.1 Scalability & Performance

ID	Requirement	Description	Priority
NFR-P3-SCAL-001	Network Scale	The architecture shall scale horizontally to support 1,000+ outlets and 10,000+ concurrent users.	High
NFR-P3-SCAL-002	Zero-Downtime Outlet Addition	New outlets shall be added to the network without downtime to existing outlets.	Medium
NFR-P3-SCAL-003	Consolidated Query Performance	Dashboard and report queries covering up to 5 years of multi-outlet data shall return results within 5 seconds.	Medium

6.3.2 Security & Compliance

ID	Requirement	Description	Priority
NFR-P3-SEC-001	Encryption	The system shall encrypt data at rest and in transit using industry-standard algorithms (AES-256, TLS 1.2+).	High
NFR-P3-SEC-002	PCI-DSS Compliance	Payment-card data handling shall comply with PCI-DSS requirements.	High
NFR-P3-SEC-003	Data Protection Compliance	The system shall comply with applicable data-protection regulations for customer PII across the jurisdictions it operates in.	High

6.3.3 Availability & Disaster Recovery

ID	Requirement	Description	Priority
NFR-P3-AVAIL-001	Cloud Uptime SLA	Cloud-hosted services shall guarantee 99.9% uptime, excluding scheduled maintenance.	High
NFR-P3-AVAIL-002	Backup & DR Targets	Cloud services shall support automated daily backup and disaster recovery with RPO ≤ 1 hour and RTO ≤ 4 hours.	High

6.3.4 Portability

ID	Requirement	Description	Priority
NFR-P3-PORT-001	Cross-Platform Clients	POS and mobile clients shall run on Windows, Android and iOS; the HO console shall run in standard web browsers.	Medium

7. External Interface Requirements

7.1 User Interfaces

- Phase 1–2: touch-optimized billing screen with a separate, simple Admin area for items, stock, purchasing and settings.
- Phase 3: adds a restaurant floor-plan view, KDS screen, responsive HO web console with drill-down navigation, and simplified mobile apps for captains and delivery.

7.2 Hardware Interfaces

- Phase 1: thermal receipt printer, barcode scanner, cash drawer, optional weighing scale.
- Phase 3 adds: kitchen printers, KDS displays, card/UPI payment terminals, optional biometric devices.

7.3 Software Interfaces

- Phase 1–2: local backup to file/USB and optional cloud backup over HTTPS; no mandatory third-party integrations.
- Phase 3 adds: payment gateway/UPI APIs, government tax portal APIs (e-invoice/e-way bill), accounting software connectors, delivery-aggregator APIs, SMS/WhatsApp/email gateways, e-commerce marketplace APIs.

7.4 Communication Interfaces

- HTTPS/TLS 1.2+ for all client-server and third-party API communication from Phase 3 onward.
- Store-local network (LAN/Wi-Fi) communication between POS terminals, KDS and printers within an outlet.

8. Data Requirements — Key Entities by Phase

Entities are introduced progressively; later phases add fields and relationships to earlier entities rather than replacing them.

Entity	Description	Introduced in Phase
Item / SKU	Sellable product with price, tax rate and stock quantity.	1
Sale / Bill & Line Item	Header and line record of a completed transaction.	1
Payment	Tender record(s) applied against a bill.	1
Stock Ledger	Chronological log of stock movement.	1
Tax Rate	Configured tax percentage assignable to items.	1
User (Admin/Cashier)	Login account and role.	1
Store Profile	Single record holding store name, address, tax ID.	1
Vendor / Supplier	Source of purchased goods with terms and ledger balance.	2
Purchase Order / GRN	Procurement request and corresponding goods-receipt record.	2
Batch / Expiry Record	Batch number and expiry date linked to a stock lot.	2
Outlet	Physical or franchise store location with its own configuration.	3
Franchise Agreement	Royalty and revenue-share terms linked to an outlet.	3
Table / KOT	Restaurant seating unit and kitchen order tickets raised against it.	3
Recipe / BOM	Ingredient composition of a finished menu item.	3
Customer / Loyalty Account	Buyer profile with purchase history and points/tier balance.	3
Promotion / Discount Rule	Configured condition-action rule governing discounts.	3

9. Representative Use Cases by Phase

Phase	ID	Use Case	Primary Actor	Brief Description
1	UC-01	Ring up a sale	Cashier	Scan/search items, apply discount if needed, collect payment, print receipt.
1	UC-02	Upload opening stock	Admin	Prepare a CSV/Excel file of items and quantities and upload it to initialize or refresh stock.
1	UC-03	Close the day	Cashier / Admin	Count cash, generate the daily Z-report, compare expected vs. actual cash.
2	UC-04	Raise and receive a purchase order	Admin	Create a PO to a vendor, approve it, and record goods receipt on delivery, updating stock automatically.
2	UC-05	Adjust stock for damage	Admin	Record a stock write-off with a reason, subject to approval, reducing stock on hand.
3	UC-06	Push a price update network-wide	HO Admin	Update an item's price centrally and publish it to selected outlets or the entire chain.
3	UC-07	Take a dine-in order	Waiter	Select table, add items and modifiers, send KOT to kitchen, confirm order to guest.
3	UC-08	View consolidated sales dashboard	Business Owner	Review real-time sales and top-item performance across all outlets.

10. Appendices

10.1 Requirement Priority Legend

Priority	Meaning (Within Its Own Phase)
High	Required for that phase's release to ship
Medium	Required for that phase's general availability; may fast-follow
Low	Desirable enhancement, may slip to the next phase

10.2 Migration Continuity Note

Every phase is designed so the next phase is additive rather than a rewrite: Phase 2's vendor and stock-maintenance entities attach to the same Item and Stock Ledger introduced in Phase 1, and Phase 3's multi-outlet Item, Customer and Stock entities are extensions of the single-store versions used in Phases 1–2. Store data from a Phase 1/2 pilot can be migrated into the Phase 3 model using the tooling defined in FR-P3-ADM-004.

10.3 Change History

Version	Date	Description	Author
1.0	September 2026	Initial draft SRS (single, non-phased scope)	
2.0	September 2026	Restructured into Phase 1 (uploaded-stock billing), Phase 2 (vendor/stock maintenance), Phase 3 (multi-outlet/franchise/restaurant chains)	